

CURSO: MATEMÁTICA PARA INGENIEROS 2 (100000110N).

TEMA: 3º ENTREGA DE GUÍAS RESUELTAS.

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Resolución de guías de las semanas 8, 9, 11 y 12.

RESOLUCION GUIA 8

Ejercicio 01. Determinar el área de la región delimitada externamente a la curva $r = 1 + \sin\theta$ e internamente al círculo $r = 3\sin\theta$.

Ejercicio 02. Determinar el área de la región delimitada externamente a la curva $r = 1$ e internamente a la cardioide $r = 1 - \sin\theta$.

Ejercicio 03. Determinar el área de la región delimitada por la curva $r = 5 - 5\sin\theta$.

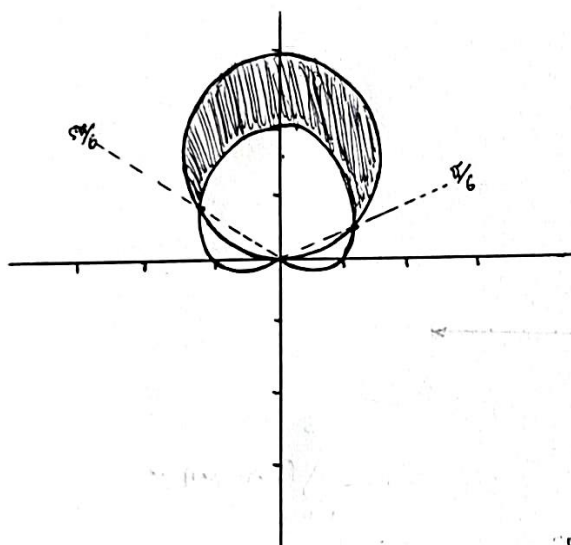
Ejercicio 04. Determinar el área de la región delimitada por la curva $r = 6 + 4\cos\theta$.

Ejercicio 05. Determinar el área de la región delimitada por la curva $r = 8\sin(3\theta)$.

Ejercicio 01.

$$r = 1 + \sin\theta$$

cardioide



$$r = 3\sin\theta$$

círculo

Paso 1

$$\text{Área} = \frac{1}{2} \int_{\pi/6}^{5\pi/6} (3\sin\theta)^2 - (1 + \sin\theta)^2 d\theta$$

$$\text{Área} = \frac{1}{2} \int_{\pi/6}^{5\pi/6} 9\sin^2\theta - (1 + 2\sin\theta + \sin^2\theta) d\theta$$

$$\text{Área} = \frac{1}{2} \int_{\pi/6}^{5\pi/6} (9\sin^2\theta - 1 - 2\sin\theta - \sin^2\theta) d\theta$$

$$\text{Área} = \frac{1}{2} \int_{\pi/6}^{5\pi/6} (8\sin^2\theta - 2\sin\theta - 1) d\theta \Rightarrow \frac{1}{2} \left[8\left(\frac{1}{2}\theta - \frac{1}{4}\sin 2\theta\right) + 2\cos\theta - \theta \right]_{\pi/6}^{5\pi/6}$$

$$\text{Área} = \frac{1}{2} \left[4\theta - 2\sin(2\theta) + 2\cos\theta - \theta \right]_{\pi/6}^{5\pi/6} \Rightarrow \frac{1}{2} \left[3\theta - 2\sin(2\theta) + 2\cos(\theta) \right]_{\pi/6}^{5\pi/6}$$

$$\text{Área} = \frac{1}{2} \left[3\left(\frac{5\pi}{6}\right) - 2\sin\left(\frac{5\pi}{3}\right) + 2\cos\left(\frac{5\pi}{6}\right) - \left(3\left(\frac{\pi}{6}\right) - 2\sin\left(\frac{\pi}{3}\right) + 2\cos\left(\frac{\pi}{6}\right)\right) \right]$$

$$\text{Área} = \frac{1}{2} \left[\frac{5\pi}{2} - \frac{\pi}{2} \right]$$

$$\text{Área} = \pi \text{ u}^2$$

Paso 2.

$$1 + \sin(\theta) = 3\sin(\theta)$$

$$1 = 3\sin(\theta) - \sin(\theta)$$

$$1 = 2\sin(\theta)$$

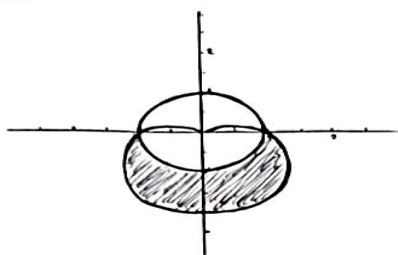
$$\frac{1}{2} = \sin(\theta)$$

$$\theta = \text{ArcSen}\left(\frac{1}{2}\right) \Rightarrow \theta = \frac{\pi}{6} \wedge \theta = \frac{5\pi}{6} \Rightarrow \left(\frac{\pi}{6} \leq \theta \leq \frac{5\pi}{6}\right)$$

Ejercicio 02
Resol

$r=1$
círculo

$r=1-\sin\theta$
cardioides



Paso 2.

$$1 - \sin\theta = 1$$

$$\sin\theta = 0$$

$$\theta = \text{Arcsen}(0)$$

$$\theta = 0 \quad \wedge \quad \theta = \pi$$

$$(0 \leq \theta \leq \pi)$$

Paso 3.

$$\text{Área} = \frac{1}{2} \int_0^\pi (1^2 - (1 - \sin\theta)^2) d\theta \Rightarrow \frac{1}{2} \int_0^\pi (1 - (1 - 2\sin\theta + \sin^2\theta)) d\theta$$

$$\text{Área} = \frac{1}{2} \int_0^\pi (1 - 1 + 2\sin\theta - \sin^2\theta) d\theta \Rightarrow \frac{1}{2} \int_0^\pi (2\sin\theta - \sin^2\theta) d\theta$$

$$\text{Área} = \frac{1}{2} \left[-2\cos(\theta) - \left(\frac{1}{4}\theta - \frac{1}{2}\sin(2\theta) \right) \right]_0^\pi \Rightarrow \frac{1}{2} \left[-2\cos(\theta) - \frac{1}{4}\theta + \frac{1}{2}\sin(2\theta) \right]_0^\pi$$

$$\text{Área} = \frac{1}{2} \left[-2\cos(\pi) - \frac{1}{4}(\pi) + \frac{1}{2}\sin(2\pi) - \left(-2\cos(0) - \frac{1}{4}(0) + \frac{1}{2}\sin(2\cdot 0) \right) \right]$$

$$\text{Área} = \frac{1}{2} \left[2 - \frac{\pi}{4} + 0 - (-2) \right]$$

$$\text{Área} = \frac{1}{2} \left[2 - \frac{\pi}{4} + 2 \right] \Rightarrow \frac{1}{2} \left[4 - \frac{\pi}{4} \right] \Rightarrow \frac{1}{2} \left(\frac{16 - \pi}{4} \right)$$

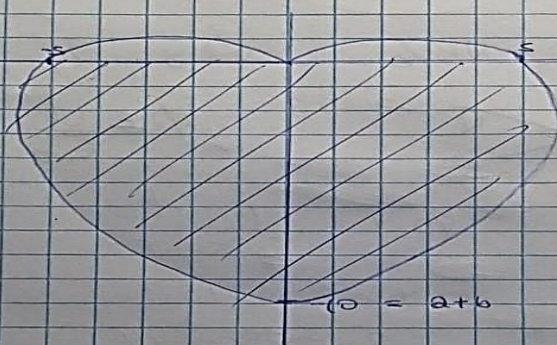
$$\text{Área} = \frac{16 - \pi}{8} \Rightarrow 1,61 \text{ u}^2$$

3.

$$\int_{\alpha}^{\beta} (a - b \sin\theta)$$

1^{er} Paso:

$$\frac{5 - 5 \sin\theta}{1 - 0} ; a = b \rightarrow \text{cardioides}$$



2^{do} Paso:

$$0 \leq \theta \leq 2\pi$$

3^{er} Paso:

$$A = \frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta$$

$$A = \frac{1}{2} \int_0^{2\pi} (5 - 5 \sin\theta)^2 d\theta$$

$$A = \frac{1}{2} \int_0^{2\pi} (25 - 50 \sin\theta + 25 \sin^2\theta) d\theta$$

$$A = \frac{1}{2} \left[25\theta - 50(-\cos\theta) + \left(\frac{1}{2}\theta - \frac{\sin 2\theta}{4} \right) \right]_0^{2\pi}$$

$$= \frac{1}{2} \left[25\theta + 50 \cos\theta + \frac{\theta}{2} - \frac{\sin 2\theta}{4} \right]_0^{2\pi}$$

$$= \frac{1}{2} \left[\frac{51\theta}{2} + 50 \cos\theta - \frac{\sin 2\theta}{4} \right]_0^{2\pi}$$

$$A = \frac{1}{2} \left[\frac{51(2\pi)}{2} + 50 \cos 2\pi - \frac{\sin 4\pi}{4} - \frac{51(0)}{2} - 50 \cos(0) + \frac{\sin 2(0)}{4} \right]$$

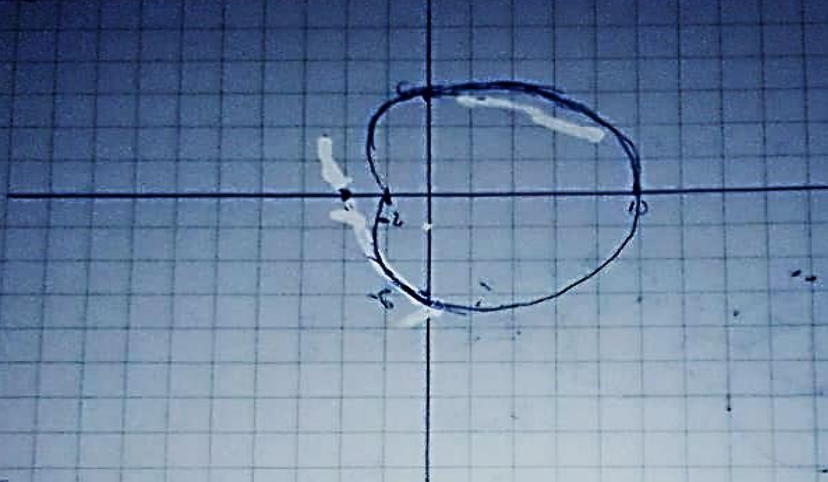
$$A = \frac{1}{2} [51\pi + 50(1) - 0/4 - 0/2 - 50(1) + 0/4]$$

$$A = \frac{51\pi}{2} = 25.5\pi$$

Definición: el área de la región del círculo que está en el primer cuadrante es:

Paso ①

Gráfico de la curva



Paso ②

Definir el intervalo de integración

Del gráfico se infiere que $0 \leq \theta \leq 2\pi$

Paso ③ Calcular el área de la región usando la fórmula:

$$\text{Área} = \frac{1}{2} \int_0^{2\pi} (6 + 4 \cos \theta)^2 d\theta$$

$$\text{Área} = \frac{1}{2} \int_0^{2\pi} (36 + 48 \cos \theta + 4 \cos^2 \theta) d\theta$$

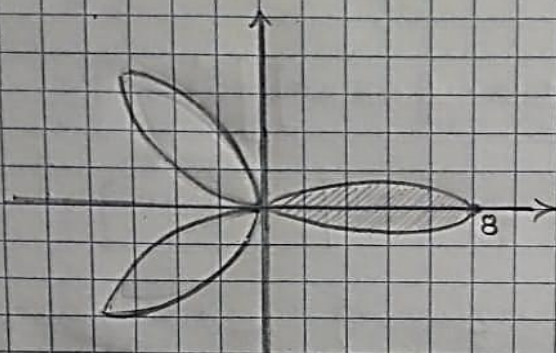
$$\text{Área} = \frac{1}{2} \left[36\theta + 48 \sin \theta + 4 \left(\frac{1}{2} \theta + \frac{1}{4} \sin 2\theta \right) \right]_0^{2\pi}$$

$$\text{Área} = \frac{1}{2} \left[36(2\pi) + 48 \sin(2\pi) + 4 \left(\frac{1}{2}(2\pi) + \frac{1}{4} \sin 2(2\pi) - 0 \right) \right]$$

$$\text{Área} = 44\pi \text{ u}^2$$

⑤ Determinar el área de la región delimitada por la curva $r = 8 \cos 3\theta$

Paso 1:



Paso 2: Definir el intervalo de integración de la Región acotada.

$$180^\circ \div 3 = 60^\circ \rightarrow \underbrace{-30^\circ \text{ a } 30^\circ}_{-\frac{\pi}{6} \text{ hasta } \frac{\pi}{6}}$$

$$-\pi/6 \leq \theta \leq \pi/6$$

Paso 3: Calcular el Área.

$$\text{Área} = \frac{1}{2} \int_{-\pi/6}^{\pi/6} (8 \cos 3\theta)^2 d\theta = \frac{1}{2} (2) \int_0^{\pi/6} (8 \cos 3\theta)^2 d\theta =$$

$$\text{Área} = \int_0^{\pi/6} (8 \cos 3\theta)^2 d\theta = 64 \int_0^{\pi/6} \cos^2 3\theta d\theta$$

$$u = 3\theta \rightarrow du = 3d\theta \rightarrow d\theta = \frac{du}{3}$$

$$\theta = \pi/6 \rightarrow u = 3(\pi/6) \rightarrow u = \pi/2$$

$$\theta = 0 \rightarrow u = 3(0) \rightarrow u = 0$$

$$\text{Área} = 64 \int_0^{\pi/2} \cos^2(u) \left(\frac{du}{3}\right)$$

$$= \frac{64}{3} \left[\frac{1}{2} u - \frac{1}{4} \sin 2u \right]_0^{\pi/2}$$

$$\text{Área} = \frac{64}{3} \left[\left(\frac{1}{2} \left(\frac{\pi}{2} \right) - \frac{1}{4} \sin 2 \left(\frac{\pi}{2} \right) \right) - \left(\frac{1}{2} (0) - \frac{1}{4} \sin 2(0) \right) \right]$$

$$\text{Área} = \frac{64}{3} \left[\left(\frac{\pi}{4} - \frac{1}{4} \sin \pi \right) \right]$$

$$\text{Área} = \frac{64}{3} \left[-\frac{\pi}{4} \right] = -\frac{16\pi}{3} \text{ u}^2$$

RESOLUCION GUIA 9

Ejercicio 01. Determinar la longitud de arco de la curva $f(x) = \frac{1}{6}x^2$ en el intervalo $0 \leq x \leq 4$. Aproximar el resultado a dos cifras decimales. Sugerencia: Usar sustitución trigonométrica para la resolución.

Ejercicio 02. Determinar la longitud de arco de la curva $f(x) = x^{\frac{3}{2}}$ en el intervalo $0 \leq x \leq 4$. Aproximar el resultado a dos cifras decimales.

Ejercicio 03. Determinar la longitud de arco de la curva $f(x) = \frac{1}{8}(x^4 + 2x^{-2})$ en el intervalo $1 \leq x \leq 2$. Aproximar el resultado a dos cifras decimales

Ejercicio 04. Determinar la longitud de arco de la curva $f(x) = -\frac{1}{2}x^2 + 2x$ en el intervalo $1 \leq x \leq 2$. Aproximar el resultado a dos cifras decimales

Ejercicio 05. Determinar la longitud de arco de la curva $f(x) = \ln(1 - x^2)$ en el intervalo $0 \leq x \leq 0,5$. Aproximar el resultado a dos cifras decimales

① $f(x) = \frac{1}{6}x^2$
 Paso ① $f'(x) = \frac{1}{3}x$
 Paso ② $[f'(x)]^2 = \left[\frac{1}{3}x\right]^2 \Rightarrow \frac{x^2}{9}$
 Paso ③ Intervalo de integración
 $0 \leq x \leq 4$
 Paso ④ Integrar
 $L = \int_0^4 \sqrt{1 + \frac{x^2}{9}} dx$
 $L = \int_0^4 \sqrt{\frac{9 + x^2}{9}} dx$
 $L = \frac{1}{3} \int_0^4 \sqrt{9 + x^2} dx$
 $L = \frac{1}{3} \left[\frac{1}{18} x \sqrt{9 + x^2} + \frac{1}{2} \ln \left(\frac{x + \sqrt{9 + x^2}}{3} \right) \right]_0^4$
 $L = \frac{1}{3} [1.66 - 0]$
 $L = 4.98$

② $f(x) = x^{3/2}$

$$f(x) = \frac{3}{2} x^{3/2 - 1}$$

$$f(x)' = \frac{3}{2} x^{1/2} \quad \leftarrow \text{Paso 1.}$$

$$f(x)' = \frac{3}{2} \sqrt{x}$$

$$[f(x)']^2 = \left(\frac{3}{2}\right)^2 \sqrt{x}^2 \quad \leftarrow \text{Paso 2.}$$

$$[f(x)']^2 = \frac{9}{4} x$$

$$0 \leq x \leq 4 \quad \leftarrow \text{Paso 3.}$$

$$L = \int_0^4 \sqrt{1 + \frac{9}{4}x} \, dx \quad \leftarrow \text{Paso 4.} \quad L = \frac{8\sqrt{(1+9/4(4))^3}}{27} - \frac{8\sqrt{(1+9/4(0))^3}}{27}$$

$$L = \int_0^4 \left(1 + \frac{9}{4}x\right)^{1/2} dx \quad L = \frac{8\sqrt{(1+27/4)^3}}{27} - \frac{8\sqrt{(1)^3}}{27}$$

$$L = \frac{4}{9} \int_0^4 \left(1 + \frac{9}{4}x\right)^{1/2} dx \quad L = 6,09$$

$$L = \frac{4}{9} \frac{(1 + 9/4 x)^{1/2+1}}{1/2+1} \Big|_0^4$$

$$L = \frac{4}{9} \frac{(1 + 9/4 x)^{3/2}}{3/2} \Big|_0^4$$

$$L = \frac{4}{9} \frac{2\sqrt{(1 + 9/4 x)^3}}{3} \Big|_0^4$$

$$L = \frac{8\sqrt{(1 + 9/4 x)^3}}{27} \Big|_0^4$$

$$f(x) = \frac{1}{8} (x^4 + 2x^{-2}) \Rightarrow f(x) = \frac{1}{8} x^4 + \frac{1}{4} x^{-2}$$

Paso 1) Derivar

$$f'(x) = \frac{x^3}{2} - \frac{x^{-3}}{2} = \frac{x^3}{2} - \frac{1}{2x^3} = \frac{2x^6 - 2}{4x^3}$$

Paso 2) $[f'(x)]^2$

$$\left[\frac{2x^6 - 2}{4x^3} \right]^2 = \frac{4x^{12} - 8x^6 + 4}{16x^6}$$

Paso 3) Intervalo $1 \leq x \leq 2$

Paso 4)

$$\int_1^2 \sqrt{1 + \frac{4x^{12} - 8x^6 + 4}{16x^6}}$$

$$\int_1^2 \sqrt{\frac{16x^6 + 4x^{12} - 8x^6 + 4}{16x^6}}$$

$$\int_1^2 \sqrt{\frac{4x^{12} + 4}{16x^6}}$$

$$\int_1^2 \frac{1}{4x^3} \sqrt{(2x^6 + 2)^2}$$

$$\int_1^2 \frac{2x^6 + 2}{4x^3}$$

$$\frac{1}{4} \int_1^2 \frac{2x^6 + 2}{x^3}$$

$$\left[\frac{1}{4} \left[\frac{x^4}{2} - \frac{1}{x^2} \right] \right]_1^2$$

$$L = \frac{1}{4} \left(\left[\frac{2^4}{2} - \frac{1}{2^2} \right] - \left[\frac{1^4}{2} - \frac{1}{1^2} \right] \right)$$

$$L = \frac{1}{4} \left(\frac{33}{4} \right)$$

$$L = \frac{33}{16} \cup \blacksquare$$

1) Determinar la longitud de arco de la curva $f(x) = -\frac{1}{2}x^2 + 2x$ en el intervalo $1 \leq x \leq 2$

Paso ① $f'(x) = -x + 2$

Paso ② $[f'(x)]^2 = (-x+2)^2$
 $= -x^2 + 4x + 4$

Paso ③ Intervalo de integración
 $1 \leq x \leq 2$

Paso ④ Integrar

$$L = \int_1^2 \sqrt{1 + (-x^2 + 4x + 4)} dx$$

$$L = \int_1^2 \sqrt{-x^2 + 4x + 5} dx$$

$$L = \int_1^2 \sqrt{-(x-2)^2 + 9} dx$$

$$L = \int_{-1}^0 \sqrt{-u^2 + 9} du$$

$$L = \int_{-1}^0 \sqrt{9 - u^2} du$$

Como de variable

$$u = x - 2$$

$$du = dx$$

$$du = dx$$

Cambio de intervalos

$$x = 2 \rightarrow x = 0$$

$$x = 1 \rightarrow x = -1$$

$$L = \left[\frac{u}{2} \sqrt{9 - u^2} + \frac{9}{2} \sin^{-1} \frac{u}{3} \right]_{-1}^0 = 0 - [-89.03] \\ L = 89.03$$

Ejercicio 05. $f(x) = \ln(1-x^2)$

Intervalo
 $0 \leq x \leq 0,5$

Resol.

$$f'(x) = \frac{1}{(1-x^2)} (-2x)$$

$$f'(x) = \frac{-2x}{1-x^2}$$

Paso 2.

$$[f'(x)]^2 = \left(\frac{-2x}{1-x^2}\right)^2 \Rightarrow \frac{4x^2}{(1-x^2)^2} \Rightarrow \frac{4x^2}{1-2x^2+x^4}$$

Paso 3.

$$L = \int_0^{0,5} \sqrt{\frac{1+4x^2}{1-2x^2+x^4}} dx$$

$$L = \int_0^{0,5} \sqrt{\frac{1-2x^2+x^4+4x^2}{(1-x^2)^2}} dx$$

$$L = \int_0^{0,5} \frac{1}{1-x^2} \sqrt{1+2x^2+x^4} dx$$

$$L = \int_0^{0,5} \frac{1}{1-x^2} \sqrt{(1+x^2)^2} dx$$

$$L = \int_0^{0,5} \frac{1+x^2}{1-x^2} dx$$

$$L = \int_0^{0,5} \left(\frac{2}{1-x^2} - 1\right) dx$$

$$L = 0,5986$$

RESOLUCION GUIA 11

Ejercicio 01. Resolver:

$$I = \int_0^1 \int_{-3}^3 \frac{xy^2}{x^2+1} dy dx$$

(Stewart, 2018, p.999)

Ejercicio 02. Resolver:

$$I = \int_0^{\frac{\pi}{6}} \int_0^{\frac{\pi}{3}} x \cdot \sin(x+y) dy dx$$

(Stewart, 2018, p.999)

Ejercicio 03. Resolver:

$$I = \int_0^1 \int_0^1 \frac{x}{1+xy} dy dx$$

(Stewart, 2018, p.999)

Ejercicio 04. Resolver:

$$I = \int_1^3 \int_1^2 \frac{1}{1+x+y} dy dx$$

(Stewart, 2018, p.999)

Ejercicio 05. Determinar el volumen del sólido delimitado por la función

$$f(x,y) = 1 - \frac{1}{4}x^2 - \frac{1}{9}y^2$$

y los planos $x = -2$, $x = 2$, $y = -1$ y $y = 1$.

(Stewart, 2018, p.999)

Ejercicio 1.

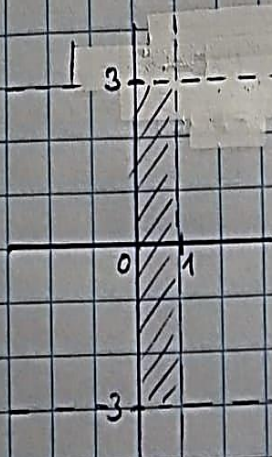
$$I = \int_0^1 \int_{-3}^3 \frac{xy^2}{x^2+1} dy dx$$

$$0 \leq x \leq 1$$

$$-3 \leq y \leq 3$$

$$I = \int_{-3}^3 \int_0^1 \frac{xy^2}{x^2+1} dx dy$$

2° grafica



3° Integramos x

$$I_x = \int_0^1 \frac{xy^2}{x^2+1} dx$$

$$I_x = y^2 \int_0^1 x \cdot \frac{1}{x^2+1} dx$$

$$u = x^2 + 1$$

$$du = 2x dx$$

$$\frac{du}{2x} = dx$$

$$\begin{cases} u=1 \Rightarrow 1^2+1=2 \\ u=0 \Rightarrow 0^2+1=1 \end{cases}$$

4. Integramos y

$$I = \int_{-3}^3 \frac{y^2}{2} [\ln 2] dy$$

$$I_x = y^2 \int_1^2 \frac{x}{u^2} du \Rightarrow \frac{y^2}{2} \int_1^2 \frac{1}{u} du$$

$$I = \frac{\ln 2}{2} \int_{-3}^3 y^2 dy$$

$$I_x = \frac{y^2}{2} [\ln u]_1^2 = \frac{y^2}{2} [\ln 2]$$

$$= \frac{\ln 2}{2} \left[\frac{y^3}{3} \right]_{-3}^3 = \frac{\ln 2}{2} \left[\frac{27}{3} - \left(-\frac{27}{3} \right) \right]$$

$$= \frac{\ln 2}{2} \cdot 18 = 9 \ln(2)$$

$$I = 9 \ln(2) \text{ ó } 6,24 u^3$$

2) $\int_0^{\pi/6} \int_0^{\pi/3} x \sin(x+y) dy dx$

$$\int u dv = uv - \int v du$$

$$I = \int_0^{\pi/3} \left[\int_0^{\pi/6} x \sin(x+y) dx \right] dy$$

$$\begin{aligned} u &= x \\ du &= dx \end{aligned} \quad \begin{cases} dv = \sin(x+y) \\ v = -\cos(x+y) \end{cases}$$

$$I = \int_0^{\pi/3} \left[-x \cos(x+y) \Big|_0^{\pi/6} + \int_0^{\pi/6} \cos(x+y) dx \right] dy$$

$$I = \int_0^{\pi/3} \left[-x \cos(x+y) \Big|_0^{\pi/6} + \sin(x+y) \Big|_0^{\pi/6} \right] dy$$

$$I = \int_0^{\pi/3} \left[-\frac{\pi}{6} \cos\left(\frac{\pi}{6}+y\right) + 0 + \sin\left(\frac{\pi}{6}+y\right) - \sin y \right] dy$$

$$I = \int_0^{\pi/3} \left[-\frac{\pi}{6} \cos\left(\frac{\pi}{6}+y\right) + \sin\left(\frac{\pi}{6}+y\right) - \sin y \right] dy$$

$$I = \int_0^{\pi/3} \left[-\frac{\pi}{6} \sin\left(\frac{\pi}{6}+y\right) - \cos\left(\frac{\pi}{6}+y\right) + \cos y \right] dy$$

$$I = \left(-\frac{\pi}{6} \sin\left(\frac{\pi}{6}+\frac{\pi}{3}\right) - \cos\left(\frac{\pi}{6}+\frac{\pi}{3}\right) + \cos \frac{\pi}{3} \right) - \left(-\frac{\pi}{6} \sin \frac{\pi}{6} - \cos \frac{\pi}{6} + 1 \right)$$

$$I = \left(-\frac{\pi}{6} + \frac{1}{2} \right) - \left(-\frac{\pi}{6} \cdot \frac{1}{2} - \frac{\sqrt{3}}{2} + 1 \right)$$

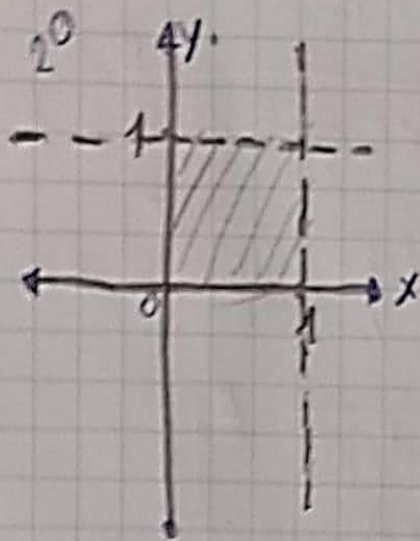
$$I = 0,104 \text{ m}^3$$

$$③ I = \int_0^1 \int_0^1 \frac{x}{1+xy} dy dx$$

$$0 \leq x \leq 1$$

$$0 \leq y \leq 1$$

$$\int_0^1 \frac{x}{1+xy} dy$$



Integrando y

$$3^\circ \quad \begin{aligned} 1+xy &= u \\ x dy &= du \\ dy &= \frac{du}{x} \end{aligned}$$

$$\int_0^1 \frac{x}{u} \frac{du}{x} = \int_0^1 \frac{1}{u} du$$

$$\Rightarrow [\ln u]_0^1 = [\ln(1+xy)]_0^1$$

$$\Rightarrow (\ln(1+x))$$

Integrando x:

$$I = \int_0^1 \ln(1+x) dx$$

$$I = [(1+x) \ln(1+x) - x]_0^1$$

$$I = 0,39 \quad u^3$$

Ejercicio 14

Resolver:

$$I = \int_1^3 \int_1^2 \frac{1}{1+x+y} dy dx$$

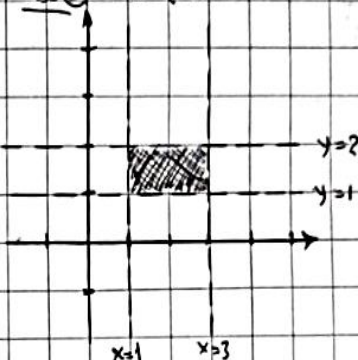
Paso 1

$$f(x,y) = \frac{1}{1+x+y}$$

$$1 \leq x \leq 3$$

$$1 \leq y \leq 2$$

Paso 2 Graficar el dominio



Paso 3

Calcular dy dx

$$I = \int_1^3 \left[\int_1^2 \frac{1}{1+x+y} dy \right] dx$$

* Reemplazamos en la segunda integral.

$$I = \int_1^3 (\ln(x+3) - \ln(x+2)) dx$$

$$I = \int_1^3 \ln(x+3) dx - \int_1^3 \ln(x+2) dx$$

$$* \int_1^3 \ln(x+3) dx$$

cambio de variable

$$u = x+3$$

$$du = dx$$

cambio de intervalo

$$\begin{cases} x=1 \rightarrow u=4 \\ x=3 \rightarrow u=6 \end{cases}$$

$$\text{Entonces,}$$

$$* \int_4^6 \ln(u) du$$

$$\Rightarrow [u \ln(u) - u]_4^6$$

$$\Rightarrow [6 \ln(6) - 4 \ln(4) - 2]$$

Finalmente

$$I = [6 \ln(6) - 4 \ln(4) - 2 - (5 \ln(5) - 3 \ln(3) - 2)]$$

$$I = 0,454$$

$$* \int_1^2 \frac{1}{1+x+y} dy$$

Cambio de variable.

$$1+x+y = u$$

$$u = 1+x+y$$

$$du = dy$$

$$\ln(u) = \ln(1+x+y)$$

$$\Rightarrow \int_1^2 \frac{1}{1+x+y} dy \Rightarrow [\ln(1+x+y)]_1^2$$

$$\Rightarrow [\ln(1+x+2) - \ln(1+x+1)]$$

$$\Rightarrow [\ln(x+3) - \ln(x+2)]$$

$$* \int_1^3 \ln(x+2) dx$$

cambio de variable

$$u = x+2$$

$$du = dx$$

cambio de intervalo

$$\begin{cases} x=1 \rightarrow u=3 \\ x=3 \rightarrow u=5 \end{cases}$$

$$\text{Entonces,}$$

$$* \int_3^5 \ln(u) du$$

$$\Rightarrow [u \ln(u) - u]_3^5$$

$$\Rightarrow [5 \ln(5) - 3 \ln(3) - 2]$$



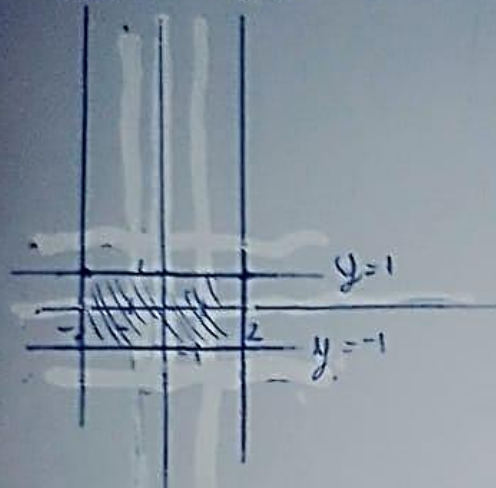
$$③ \quad f(x,y) = 1 - \frac{1}{4}x^2 - \frac{1}{9}y^2$$

Paso ① Intervalos

$$-2 \leq x \leq 2$$

$$-1 \leq y \leq 1$$

Paso ② $x = -2$ $x = 2$



Paso ③ Coso: $dy \, dx$

$$\int_{-2}^2 \left[\int_{-1}^1 1 - \frac{1}{4}x^2 - \frac{1}{9}y^2 \, dy \right] dx$$

$$I = \int_{-1}^1 1 - \frac{1}{4}x^2 - \frac{1}{9}y^2 \, dy$$

$$I = \left[y - \frac{1}{4}x^2 y - \frac{y^3}{27} \right]_{-1}^1 \Rightarrow \left[1 - \frac{1}{4}x^2(1) - \frac{(1)^3}{27} \right] - \left[-1 - \frac{1}{4}x^2(-1) - \frac{(-1)^3}{27} \right]$$

$$1 - \frac{1}{4}x^2 - \frac{1}{27} + 1 - \frac{1}{4}x^2 - \frac{1}{27}$$

$$\frac{52}{27} - \frac{1}{2}x^2$$

$$\begin{aligned}
 I &= \int_{-2}^2 \frac{52}{27} - \frac{1}{2} x^2 dx \\
 I &= \left[\frac{52}{27} x - \frac{x^3}{6} \right]_{-2}^2 \\
 I &= \left[\frac{52}{27} (2) - \frac{(2)^3}{6} \right] - \left[\frac{52}{27} (-2) - \frac{(-2)^3}{6} \right] \\
 I &= \frac{104}{27} - \frac{8}{6} + \frac{104}{27} - \frac{8}{6} \\
 I &= \frac{136}{27} m^3
 \end{aligned}$$

RESOLUCION GUIA 12

Grupo de preguntas 02. Ejercicios para los estudiantes.

Ejercicio 01. Calcular el volumen del cuerpo delimitado por el paraboloide circular $z = 1 + x^2 + y^2$ y la región bidimensional $R = \{(x; y) \in \mathbb{R}^2 : x = 0; y = 0; x + y = 1\}$.

(Espinoza, 2012, p.614)

Ejercicio 02. Determinar la integral doble, según la región de dominio R .

$$I = \iint \frac{x^2}{1+y^2} dA \quad ; \quad R = [0; 1] \times [0; 1]$$

(Espinoza, 2012, p.593)

Ejercicio 03. Determinar la integral doble, según la región de dominio R .

$$I = \iint 2x dA \quad ; \quad R = \{(x; y) \in \mathbb{R}^2 : 4y = x^2; x - 2y + 4 = 0\}$$

(Espinoza, 2012, p.594)

Ejercicio 04. Determinar la integral doble, según la región de dominio R .

$$I = \iint x dA \quad ; \quad R = \{(x; y) \in \mathbb{R}^2 : x^2 + y^2 = 4; 0 \leq x; y \leq 0\}$$

(Espinoza, 2012, p.595)

Ejercicio 05. Determinar la integral doble, según la región de dominio R .

$$I = \iint \frac{x^2}{y^2} dA \quad ; \quad R = \{(x; y) \in \mathbb{R}^2 : x = 2; y = x; xy = 1\}$$

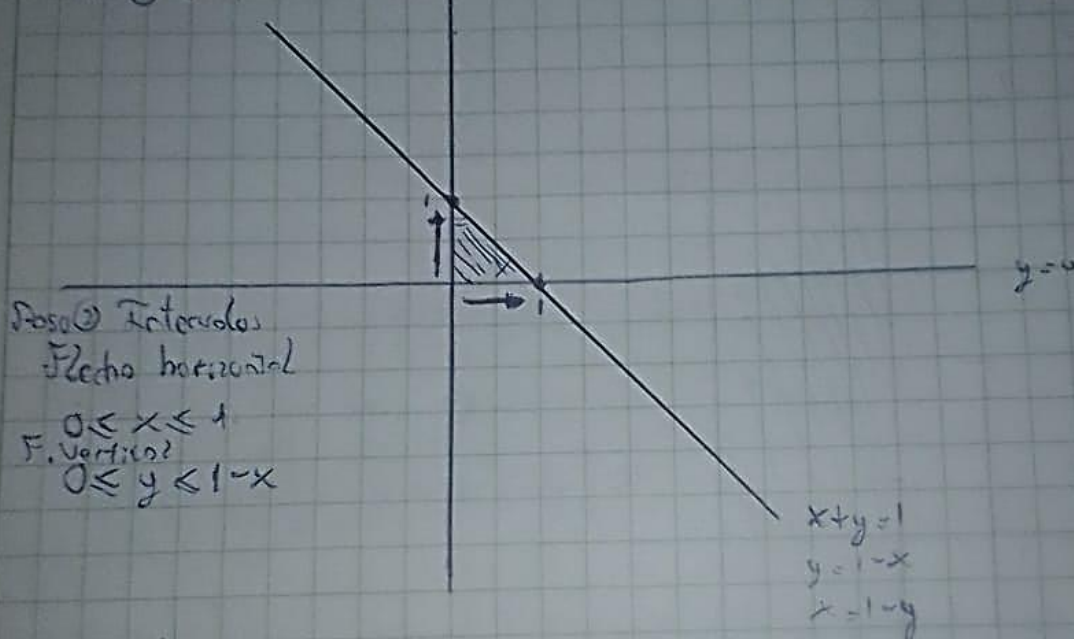
(Espinoza, 2012, p.595)

Ejercicio ①

Calcular el volumen del cuerpo delimitado por el paraboloide circular
 $z = 1 + x^2 + y^2$ y la región bidimensional $R = \{(x, y) \in \mathbb{R}^2 : x \geq 0, y \geq 0, x + y = 1\}$
 $x + y = 1$

$$f(x, y) = 1 + x^2 + y^2, x \geq 0$$

Paso ① Geometría



$$I = \int_0^1 \left(\int_0^{1-x} 1 + x^2 + y^2 \, dy \right) dx$$

$$I_1 = \int_0^{1-x} 1 + x^2 + y^2 \, dy$$

$$I_1 = \left[y + x^2 y + \frac{y^3}{3} \right]_0^{1-x}$$

$$I_1 = (1-x) + x^2(1-x) + \frac{(1-x)^3}{3}$$

$$I_1 = (1-x) + (x^2 - x^3) + \frac{(-x^3 + 3x^2 - 3x + 1)}{3}$$

$$I_1 = \frac{1}{3} (2 - 4x + 4x^2 - 2x^3)$$

$$I_1 = \frac{2}{3} (1 - 2x + 2x^2 - x^3)$$

$$I_2 = \int_0^1 \frac{2}{3} (1 - 2x + 2x^2 - x^3) dx$$

$$I_2 = \frac{2}{3} \left[x - x^2 + \frac{2x^3}{3} - \frac{x^4}{4} \right]_0^1$$

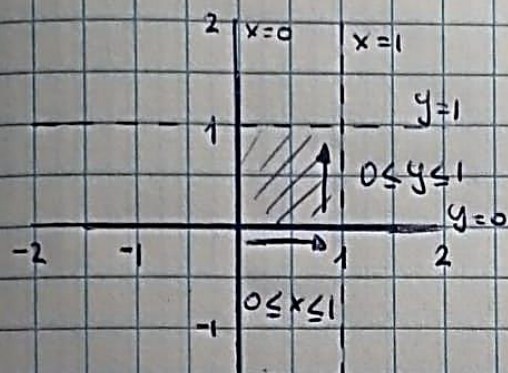
$$I_2 = \frac{2}{3} \left[\frac{5}{12} \right]$$

$$I_2 = \frac{5}{18} \text{ m}^3$$

2.

$$I = \iint \frac{x^2}{1+y^2} dA$$

$$R = [0,1] \times [0,1]$$



Caso 2:

$$I = \int_0^1 \left[\int_0^1 \frac{x^2 dy}{1+y^2} \right] dx$$

Integral Interna:

$$I = \int_0^1 \frac{x^2}{1+y^2} dy \Rightarrow x^2 \int_0^1 \frac{1}{1+y^2} dy \Rightarrow x^2 [\arctg(y)]_0^1$$

$$I = x^2 \left[\frac{1}{4}\pi - 0 \right] \Rightarrow \frac{x^2\pi}{4}$$

Integral externa:

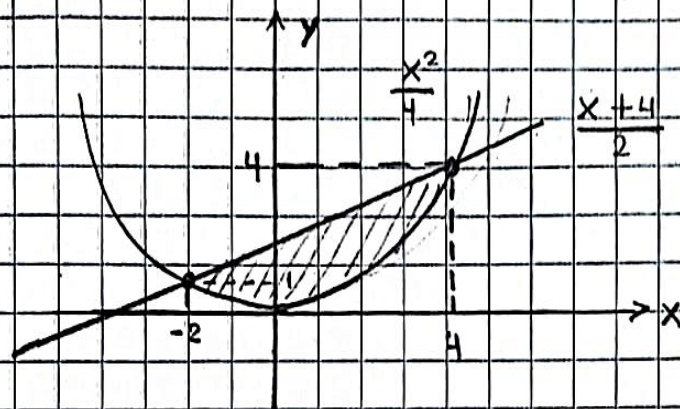
$$I = \int_0^1 \frac{x^2\pi}{4} dx \Rightarrow \frac{\pi}{4} \int_0^1 x^2 dx$$

$$I = \frac{\pi}{4} \left[\frac{x^3}{3} \right]_0^1$$

$$I = \frac{\pi}{4} \left[\frac{1}{3} - 0 \right]$$

$$I = \frac{\pi}{12}$$

③ $I = \iint_R 2x \, dA, R = \{(x,y) \in \mathbb{R}^2, 4y = x^2, x - 2y + 4 = 0\}$



$$\frac{x^2}{4} \leq y \leq \frac{x+4}{2}$$

$$I = \int_{-2}^4 \int_{\frac{x^2}{4}}^{\frac{x+4}{2}} 2x \, dy \, dx \quad I = \frac{1}{2} \left[\frac{2}{3} x^3 + 4x^2 - \frac{x^4}{4} \right]_{-2}^4$$

$$I = \int_{-2}^4 \left[\int_{\frac{x^2}{4}}^{\frac{x+4}{2}} 2x \, dy \right] dx = \left(\frac{2}{3} (4)^3 + 4(4)^2 - \frac{(4)^4}{4} \right) - \left(\frac{2}{3} (-2)^3 + 4(-2)^2 - \frac{(-2)^4}{4} \right)$$

$$I = \int_{-2}^4 \left[2xy \right]_{\frac{x^2}{4}}^{\frac{x+4}{2}} dx \quad I = \frac{1}{2} \times 36 = 18$$

$$I = \int_{-2}^4 2x \left(\frac{x+4}{2} - \frac{x^2}{4} \right) dx$$

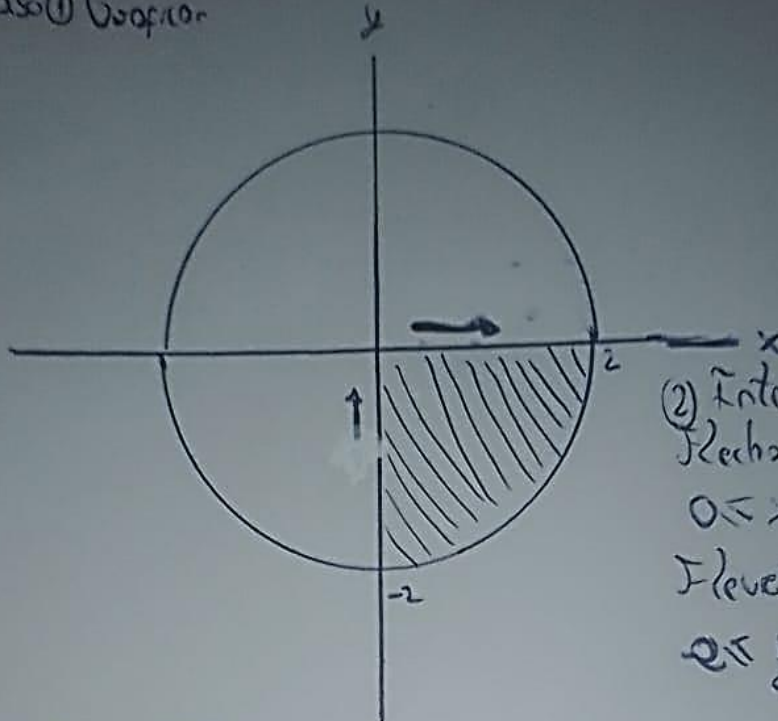
$$I = \frac{1}{2} \int_{-2}^4 4x \left(\frac{x+4}{2} - \frac{x^2}{4} \right) dx$$

$$I = \frac{1}{2} \int_{-2}^4 (2x^2 + 8x - x^3) dx$$

$$④ \quad I = \iint x \, dA$$

$$R: x^2 + y^2 = 4, \quad 0 \leq x, \quad y \leq 0$$

Paso ① Gráfico



② Intervalo.
Flecha horizontal
 $0 \leq x \leq \sqrt{4-y^2}$
Flecha vertical
 $0 \leq y \leq 0$

$$I = \int_{-2}^0 \left(\int_0^{\sqrt{4-y^2}} x \, dx \right) dy$$

$$I = \int_{-2}^0 \left[\frac{x^2}{2} \right]_0^{\sqrt{4-y^2}} dy$$

$$I = \left[\frac{x^2}{2} \right]_0^{\sqrt{4-y^2}} \Rightarrow \frac{(\sqrt{4-y^2})^2}{2} = \frac{4-y^2}{2}$$

$$I = \int_{-2}^0 \frac{4-y^2}{2} dy$$

$$I = \frac{1}{2} \left[4y - \frac{y^3}{3} \right]_{-2}^0 = \frac{1}{2} \left[4(0) - \frac{(0)^3}{3} - \left(4(-2) - \frac{(-2)^3}{3} \right) \right] = \frac{35}{6} \text{ u}^3$$

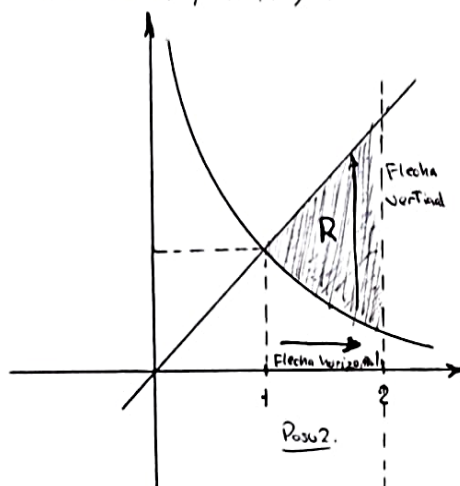
Ejercicio 05

$$I = \iint_R \frac{x^2}{y^2} dA ; R = \{(x,y) \in \mathbb{R}^2 : x=2 ; y=x ; xy=1\}$$

Paso 1. "Grificar"

$$f(x,y) = \frac{x^2}{y^2}$$

$$R : x=2 ; y=x ; xy=1$$



$$* 1 \leq x \leq 2$$

$$* \frac{1}{x} \leq y \leq x$$

Paso 3. Paso 1: $dy dx$

$$I = \int_1^2 \left[\int_{1/x}^x \frac{x^2}{y^2} dy \right] dx$$

Primera integral, integral interna.

$$* \int_{1/x}^x \frac{x^2}{y^2} dy \Rightarrow x^2 \int_{1/x}^x \frac{1}{y^2} dy \Rightarrow x^2 \int_{1/x}^x \left(\frac{1}{y}\right)^2 dy \Rightarrow x^2 \int_{1/x}^x y^{-2} dy$$

$$\Rightarrow x^2 \left[-\frac{y^{-1}}{1} \right]_{1/x}^x \Rightarrow x^2 \left[-\frac{1}{y} \right]_{1/x}^x \Rightarrow x^2 \left[-\frac{1}{x} - \left(-\frac{1}{1/x} \right) \right]$$

$$\Rightarrow x^2 \left[-\frac{1}{x} + \frac{1}{1/x} \right] \Rightarrow \frac{x^2}{1/x} - \frac{x^2}{x} \Rightarrow x^3 - x$$

* Integral Externa

$$I = \int_1^2 x^3 - x dx \Rightarrow \left[\frac{x^4}{4} - \frac{x^2}{2} \right]_1^2$$

$$I = \left[\frac{2^4}{4} - \frac{2^2}{2} \right] - \left[\frac{1^4}{4} - \frac{1^2}{2} \right]$$

$$I = 2 - \left(-\frac{1}{4} \right)$$

$$I = 2 + \frac{1}{4}$$

$$I = \frac{9}{4}$$